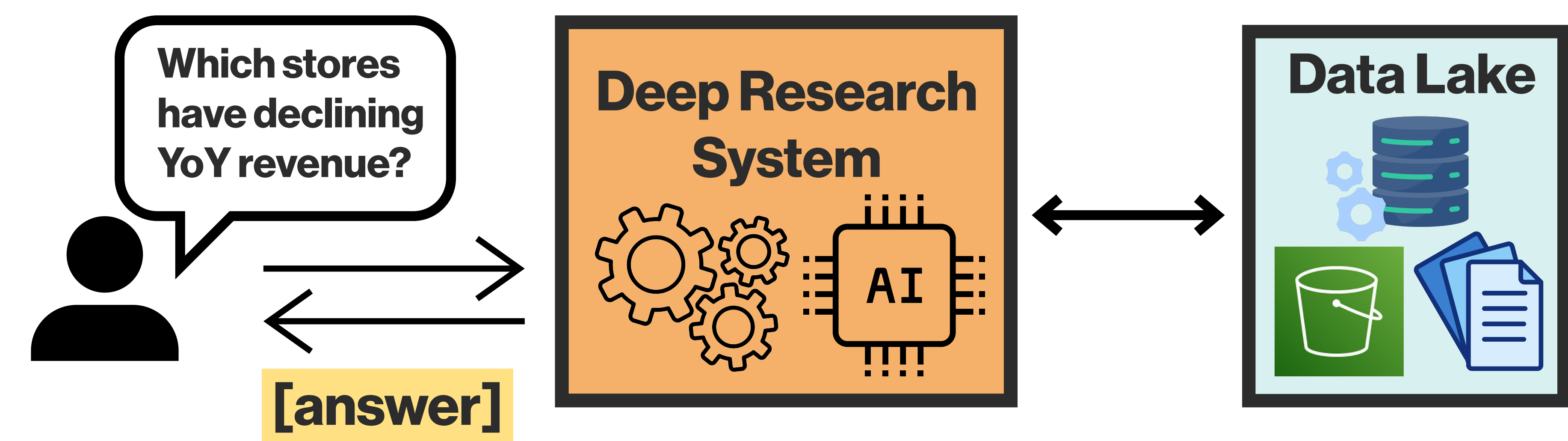


Deep Research Promises to Accelerate Productivity and Transform Information Work



Deep Research promises to simplify information work:

- 1

Connect your data lake
- 2

Ask a question in natural language
- 3

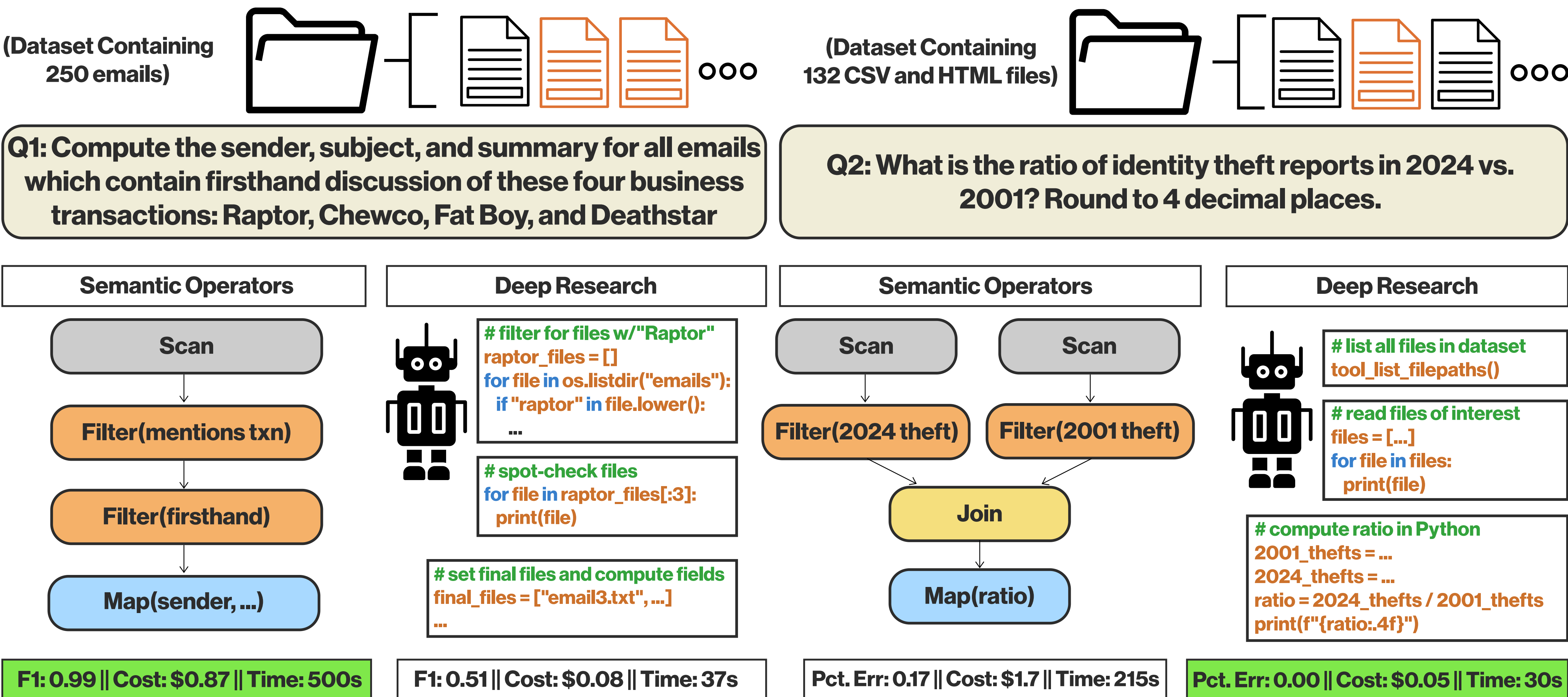
Get your answer!

- PROS

- Works on structured and unstructured data
 - Works on multiple data modalities (txt, img, video, etc.)
 - Users do not have to know schemas / SQL / where data lives within their data lake
- CONS

- Correctness is not guaranteed
 - Current systems are slow and expensive

But Can it Deliver on that Promise? Let's Consider Two Unstructured Analytics Queries



Observation: Deep Research Agents Create Good Query Plans, but can Suffer from Poor Execution

Can we Combine Optimized Execution of Semantic Operators w/Dynamic Execution of Deep Research?

	Optimized Execution	Dynamic Execution	Supports Natural Language Queries	First-Class Support of Unstructured Data
OLAP Databases	✓	✗	🤔	✗
Semantic Operator Systems	✓	✗	🤔	✓
Deep Research Systems	✗	✓	✓	✓
AI Query Engine	✓	✓	✓	✓

Early Results: Quality Improvements and Cost and Runtime Savings of >70%

Q1	Cost	Runtime	F1	Q2	Cost	Runtime	Pct. Err.
Deep Research	\$0.08	37s	51%	Semantic Operators	\$1.66	215s	17.00%
Deep Research + Sem. Ops.	\$3.76	2,000s	99%	Deep Research	\$0.03	77s	27.56%
Our Prototype	\$0.87	546s	99%	Our Prototype	\$1.17	583s	0.02%

Prototype Idea: Let's Give Deep Research Agents the Ability to Use and Optimize Semantic Operators

Our Future Work Will Expand on This Vision and Build a General-Purpose System & Optimizer for Deep Research